



OPEN T1-weighted MRI-based brain tumor classification using hybrid deep learning models

Mohsen Asghari Ilani¹, Dingjing Shi² & Yaser Mike Banad¹✉

Health is fundamental to human well-being, with brain health particularly critical for cognitive functions. Magnetic resonance imaging (MRI) serves as a cornerstone in diagnosing brain health issues, providing essential data for healthcare decisions. These images represent vast datasets that are increasingly harnessed by deep learning for high-performance image processing and classification tasks. In our study, we focus on classifying brain tumors—such as glioma, meningioma, and pituitary tumors—using the U-Net architecture applied to MRI scans. Additionally, we explore the effectiveness of convolutional neural networks including Inception-V3, EfficientNetB4, and VGG19, augmented through transfer learning techniques. Evaluation metrics such as F-score, recall, precision, and accuracy were employed to assess model performance. The U-Net segmentation architecture, emerged as the top performer, achieving an accuracy of 98.56%, along with an F-score of 99%, an area under the curve of 99.8%, and recall and precision rates of 99%. This study demonstrates the effectiveness of U-Net, a convolutional neural network architecture, for accurate brain tumor segmentation in early detection and treatment planning. Achieving an accuracy of 96.01% in cross-dataset validation with an external cohort, U-Net exhibited robust performance across diverse clinical scenarios. Our findings highlight the potential of U-Net and transfer learning in enhancing diagnostic accuracy and informing clinical decision-making in neuroimaging, ultimately improving patient care and outcomes.

Keywords Brain tumor, Medical imaging, Convolutional neural networks, U-Net, Transfer learning

Cancer arises from the uncontrolled growth and spread of abnormal cells throughout the body. In healthy individuals, cells follow a tightly regulated process of growth, division, and programmed death^{1–3}. However, disturbances in these processes can precipitate the development of tumors. Tumors may be benign or malignant, with malignant tumors invading nearby tissues and potentially spreading to other parts of the body, forming new tumors⁴. There are over 100 types of cancer, named after the tissues or organs where they originate. These cancers fall into two main categories: benign tumors, such as meningiomas and pituitary tumors, which grow slowly and are less likely to spread, and malignant tumors like astrocytomas and glioblastomas, which are cancerous, invasive, and challenging to treat due to their aggressive nature and propensity to spread within the central nervous system⁵. Our study focuses on the diagnosis of three significant brain conditions: Meningiomas, the most common primary brain tumor affecting mainly women, typically noncancerous but potentially persistent; Pituitary adenomas, prevalent among pituitary tumors, characterized by slow growth and treatable benignity; and Gliomas, including astrocytomas and glioblastomas, common brain tumors originating from glial cells, with varying aggressiveness based on genetic mutations⁶.

MRI, or magnetic resonance imaging, utilizes a powerful magnetic field to create detailed images of the brain or spinal cord without causing discomfort or harm to patients. For brain tumor patients, MRI scans are scheduled regularly as part of their medical care, with the frequency adjusted based on tumor type and clinical assessment. The technique involves aligning and disturbing the magnetization of protons in tissue water nuclei using external radio frequency (RF) energy. These nuclei emit RF signals as they return to their original state, which are captured and transformed into grayscale images using Fourier transformation. Different types of images are produced by varying the sequence of RF pulses applied and collected. MRI parameters such as Repetition Time (TR) and Time to Echo (TE) dictate the timing of pulse sequences and echo signals, respectively. Tissue characteristics are defined by T1 (longitudinal relaxation time) (Fig. 1) and T2 (transverse relaxation time), which indicate how quickly protons realign with the magnetic field and lose phase coherence, respectively^{7,8}.

Deep learning has rapidly become a cornerstone in various medical imaging tasks, offering powerful feature extraction and classification capabilities without extensive manual preprocessing or expert-driven heuristics.

¹School of Electrical and Computer Engineering, University of Oklahoma, Norman, OK 73019, USA. ²Department of Psychology, University of Oklahoma, Norman, OK 73019, USA. ✉email: bana@ou.edu

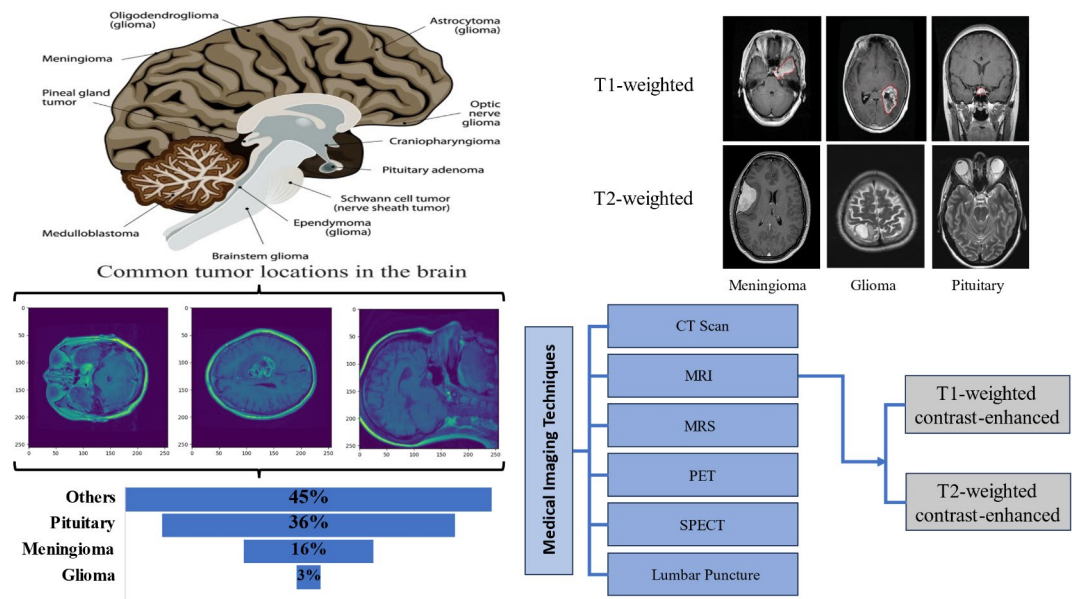


Fig. 1. Brain Tumor Types and the application of medical imaging techniques for computer-aided Diagnosis (CAD).

Recent surveys and reviews consistently highlight the transformative impact of deep learning in medical image analysis—encompassing disease detection, segmentation, and classification across imaging modalities such as MRI, CT, and ultrasound^{9–11}. Beyond these image-centric approaches, interdisciplinary insights from the control and optimization fields also inform robust design demonstrating how deep learning can address complex, dynamic environments^{12,13}. Deep learning, mimicking human intelligence, significantly enhances the accuracy of diagnosis, treatment planning, and patient management, playing a crucial role in advancing medical practices, particularly in addressing complex issues such as brain tumors. Deep learning, notably through convolutional neural networks (CNNs), has revolutionized medical imaging by automating feature extraction from images without manual intervention¹⁴. However, CNNs require substantial computational resources and extensive labeled datasets for optimal performance. Transfer learning mitigates these challenges by leveraging pre-trained models, reducing training time and enhancing efficiency in medical image analysis.

The U-Net architecture has emerged as a critical tool in medical imaging for detecting brain tumors, distinguished by its specialized design crafted for precise segmentation tasks. Unlike conventional CNNs, U-Net preserves crucial spatial information essential for accurately outlining tumor boundaries through skip connections that maintain detailed information across various scales^{15,16}. This capability is particularly crucial in biomedical imaging, where precise localization and characterization of tumors are pivotal for treatment planning and monitoring. By effectively integrating local and global context, U-Net enhances the accuracy of segmentation results, thereby improving diagnostic capabilities even with limited annotated datasets common in medical applications. Its adeptness in addressing these challenges positions U-Net as a robust support tool for clinicians in early tumor detection and comprehensive evaluation, ultimately leading to more informed medical decisions and improved patient outcomes.

This study endeavors to overcome the challenges associated with detecting, classifying, and segmenting brain tumors through the development of a sophisticated comparative system. By leveraging advanced CNN architectures and training methodologies, including U-Net, residual units, batch normalization (BN), dropout regularization, parametric rectified linear unit (PReLU), and skip connections, the system integrates both tumor context and local spatial features. Unlike conventional methods, this approach considers pixel arrangement and local label dependencies to achieve heightened classification accuracy. However, significant obstacles such as computational complexity and prolonged training times persist, underscoring the need to explore transfer learning techniques as potential solutions.

Transfer learning (TL) has garnered considerable interest in the medical domain by utilizing pretrained models to minimize the requirement for extensive training data and computational resources. Notably, models like VGG, MobileNet, Inception, ResNet, and DenseNet^{14,17–19} have demonstrated impressive effectiveness in analyzing medical images. These deep and intricate neural networks excel in identifying complex patterns inherent in medical data. The adoption of transfer learning has expedited model deployment, enhancing the accuracy and efficacy of pretrained models. This advancement enables quicker and more precise diagnoses of cancerous lesions, thereby improving patient outcomes.

In this study, we conducted a comparative analysis of the U-Net architecture, a Fully Connected CNN, and three CNN-based transfer learning models—Inception-V3, EfficientNetB4, and VGG19—to evaluate their effectiveness in classifying brain MRI data. While U-Net is traditionally designed for image segmentation, it has been increasingly adapted for classification tasks in medical imaging. Our motivation for incorporating

U-Net lies in its ability to retain spatial information through skip connections, which can enhance classification performance when properly modified. Unlike conventional CNNs that primarily focus on down-sampling, U-Net effectively captures both local and global contextual features, making it useful for medical images where subtle tumor boundaries play a crucial role in diagnosis.

To establish a baseline for comparison, we included a 3-layer CNN, inspired by the work of Mohammad Zafer Khaliki (Published: 01 February 2024)¹⁴, where a simple CNN was employed for brain tumor classification. This inclusion allows us to demonstrate the performance improvements offered by deeper architectures (transfer learning models) and segmentation-based approaches (U-Net). The baseline model provides insight into how modern deep learning techniques surpass traditional CNNs in terms of feature extraction, accuracy, and robustness.

Additionally, our research underscores the importance of transfer learning and fine-tuning techniques in MRI-based brain tumor classification, facilitating improved generalizability and clinical applicability across diverse datasets:

- (i) We meticulously optimized the U-Net segmentation architecture for three-class brain tumor classification, leveraging advanced down-sampling and up-sampling methods to improve feature extraction and overall performance.
- (ii) Following meticulous preprocessing steps, we fine-tuned state-of-the-art transfer learning models—Inception-V3, EfficientNetB4, and VGG19—augmenting them with additional fully connected layers to optimize their performance across diverse benchmark datasets.
- (iii) In establishing a robust comparative framework, we benchmarked our advanced transfer learning methodologies against previous studies, demonstrating outstanding precision metrics. Notably, U-Net models performed with accuracy of 98.56%, VGG19 achieved a historical accuracy of 97.72, while EfficientNetB4 showcased exceptional precision of 98.31% in practical applications.
- (iv) Transfer learning facilitated the utilization of pre-trained models, proving instrumental in effectively leveraging limited annotated medical data and reducing computational complexity during training.
- (v) Additionally, external cohort cross-dataset validation further underscored the efficacy of our methodologies, with U-Net maintaining robust performance, achieving an accuracy of 96.01% in diverse clinical scenarios.
- (vi) Our findings underscore the significant contribution of CNN-based transfer learning approaches in improving the accurate identification of brain tumors. These approaches demonstrate substantial promise for enhancing diagnostic precision and informing clinical decision-making processes within the field of medical imaging.

Figure 2 illustrates the study's flow and models' details in Table 1. This article unfolds as follows: We first review relevant literature (Related Work), then detail our methodology, experimental setup, and evaluation methods (Materials and Methods). Next, we present and analyze our findings (Results and Discussion), concluding with key takeaways and future research avenues (Conclusion and Future Work).

Related work

In recent decades, the field of brain tumor diagnosis has garnered significant research attention, leading to the introduction of various diagnostic techniques. Automatic brain tumor detection methods have notably enhanced diagnostic performance. Machine learning plays a pivotal role in identifying, classifying, and segmenting brain tumors, with several algorithms developed specifically for this purpose, as detailed in Table 2.

The classification of brain tumors is crucial in medical diagnostics, significantly impacting patient outcomes. Recent research has extensively explored deep learning techniques such as CNN, U-Net, and Transfer Learning-based CNNs for this purpose. Haq et al.²⁰ introduced a deep learning method to improve brain tumor classification accuracy through data augmentation and transfer learning. Abd-Allah et al.¹⁵ developed an automatic brain tumor diagnosis system using CNNs for detection, classification, and segmentation of glioblastomas, achieving up to 99% classification accuracy on the BraTS 2017 dataset. Swati et al.²¹ addressed MRI image classification challenges by proposing a block-wise fine-tuning strategy with pre-trained convolutional neural networks (CNNs), achieving an average accuracy of 94.82% under five-fold cross-validation on the CE-MRI dataset. Dong et al.²² used a U-Net model with extensive data augmentation to create a fully automatic 2D method for brain tumor segmentation. Their approach incorporated a 3D U-Net network that leveraged shortcut connections between upsampling and downsampling paths, demonstrating strong performance in tumor segmentation tasks. Another study explored various methodologies for brain tumor detection, segmentation, and classification, underscoring the versatility and effectiveness of U-Net in medical imaging applications²³.

Xu et al.²⁴ introduced Prov-GigaPath, a robust whole-slide pathology model pretrained on extensive datasets, achieving state-of-the-art results across 25 out of 26 digital pathology tasks. This underscores the critical role of comprehensive real-world data and whole-slide modeling in advancing digital pathology. Vimala et al.²⁵ explored EfficientNets for brain tumor classification, employing transfer learning-based fine-tuning to categorize glioma, meningioma, and pituitary tumors. EfficientNetB2 demonstrated strong performance with an overall test accuracy of 99.06%, precision of 98.73%, recall of 99.13%, and an F1-score of 98.79%. Khaliki et al.¹⁴ compared CNN architectures, including Inception-V3, EfficientNetB4, and VGG19, for brain tumor classification from MR images. VGG16 emerged as the top performer with 98% accuracy, 97% F-score, 99% AUC, 98% recall, and 98% precision, highlighting its efficacy in medical image analysis. Mathivanan et al.²⁶ evaluated deep transfer learning architectures such as ResNet152, VGG19, DenseNet169, and MobileNetv3 for brain tumor diagnosis, with MobileNetv3 achieving the highest accuracy at 99.75%, underscoring the transformative potential of TL in enhancing diagnostic accuracy.

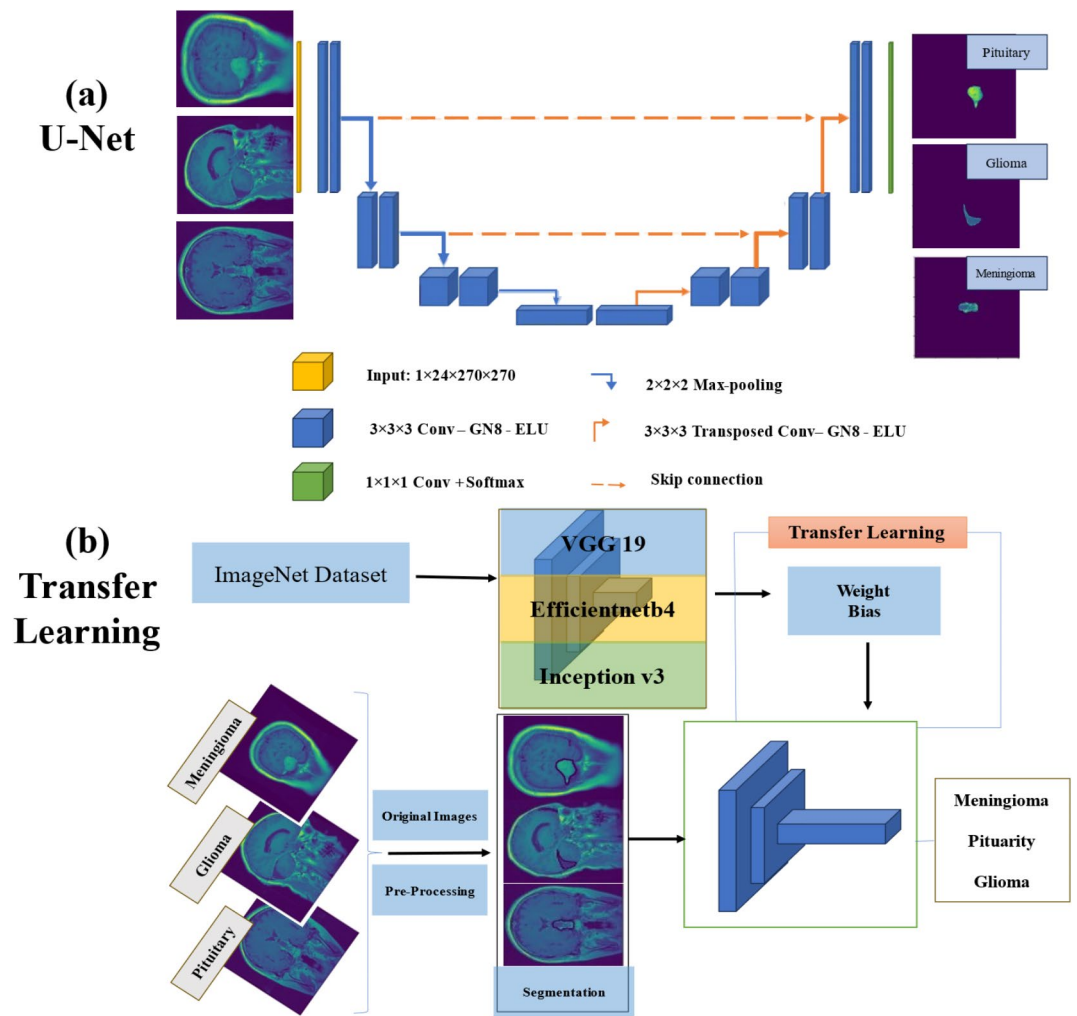


Fig. 2. The flow diagram of this study, (a) using U-Net architecture, (b) Transfer Learning-based CNN architecture.

Model	Number of layers	Kernel sizes	Pooling	Activation functions	Regularization	Optimization	Total parameters
3-Layer CNN	3 Conv + 1 Dense	3 × 3	MaxPooling	ReLU	L2 + Dropout (0.5)	Adam	~ 300 K
U-Net	Encoder-Decoder (+ 10 Layers)	3 × 3	MaxPooling + Upsampling	ReLU	BatchNorm + Dropout (0.5)	Adam	~ 31 M
VGG19	19	3 × 3	MaxPooling	ReLU	BatchNorm + L2	SGD + Momentum	~ 20 M
InceptionV3	42	1 × 1, 3 × 3, 5 × 5	Avg & Max Pooling	ReLU + Softmax	BatchNorm + Dropout	RMSProp	~ 23 M
EfficientNetB4	55	varied (3 × 3, 5 × 5)	Avg & Max Pooling	Swish + Softmax	BatchNorm + Dropout	Adam	~ 19 M

Table 1. Summary of model architectures used in brain tumor classification.

Deepak et al.²⁷ utilized deep transfer learning with a pretrained GoogLeNet to classify brain MRI images, achieving an average accuracy of 98%. This study highlights the effective application of transfer learning in medical imaging, demonstrating robust performance despite limited training data. Zulfiqar et al.²⁸ applied EfficientNets to brain tumor classification through fine-tuning, resulting in substantial performance gains. EfficientNetB2 achieved an overall test accuracy of 98.86%, with precision, recall, and F1-score metrics of 98.65%, 98.77%, and 98.71%, respectively. The study effectively employed Grad-CAM visualizations to highlight tumor regions. Samitha et al.²⁹ explored multiple EfficientNet models for brain tumor classification, identifying EfficientNetB2 as the top performer and showcasing significant accuracy improvements. Emam et al.³⁰ developed an improved residual learning architecture for classifying brain tumors, optimized using an enhanced Hunger Games Search algorithm. Their proposed I-HGS-ResNet50 model achieved exceptional accuracies of 99.89%,

References	Model	Dataset	Metrics (Accuracy)
Vimala et al. ²⁵	EfcientNetB2	Figshare	99.06%
Khaliki et al. ¹⁴	VGG16	Kaggle	98%
Mathivanan et al. ²⁶	ResNet152	Kaggle	96.92%
	VGG19		95.62%
	DenseNet169		97.53%
	MobileNetv3		98.52%
Abd-Ellah et al. ¹⁵	Proposed TRDCNN	BraTS 2017	98.88%
Haq et al. ²⁰	ResNet50	China's Nanfang hospital	97.3%
	Inception V3		93.23%
	Xception		93%
	MobilleNet		96.76%
Deepak et al. ²⁷	GoogLeNet	Figshare	97.1%

Table 2. Literature review: comparative analysis of classification and segmentation techniques.

99.72%, and 99.88% across various datasets, surpassing benchmarks set by previous studies and other prominent deep learning models.

Materials and methods

This section details the dataset, and methodologies used in our study, focusing on the classification of three brain tumor types: meningioma with 708 MRI slices, glioma with 1426 slices, and pituitary tumor with 930 slices. The study utilized a basic fully connected CNN and U-Net architectures for classification tasks. Furthermore, transfer learning techniques were applied, leveraging pre-trained models such as VGG19, InceptionV3, and EfficientNetB4, to improve classification accuracy.

Data source

The Figshare dataset, originally introduced by Cheng in 2015 and updated in 2024, is publicly available as a repository³¹. The dataset consists of 3064 T1-weighted contrast-enhanced MRI images of the human brain, categorized into three classes: meningioma (class 0), glioma (class 1), and pituitary tumor (class 2).

Pre-processing

Initially, the input images are resized from 512 × 512 × 3 to appropriate dimensions to ensure efficient processing by the models. These resizing preserves image content and features while optimizing computational efficiency during network training.

Figure 3 illustrates the MR image cropping procedure by removing the outer edges of the brain contour. The process begins with converting images to grayscale, followed by Gaussian blur filtering. Thresholding is then applied, and subsequent erosion and dilation operations refine the image by reducing noise. The largest contours are identified from the thresholded image, determining the corners (top, bottom, left, and right) of the original image. This method effectively removes extraneous image data, enhancing the clarity and quality of the training dataset.

To ensure robust training and prevent bias, the MRI scans of brain tumors are randomly shuffled before processing. After cropping to remove irrelevant sections, the dataset is split into an 80:20 ratio for training and testing. Specifically, 70% out of 80% is allocated for training, while the remaining 30% is used as a validation set to evaluate model performance on unseen samples. Table 3 shows the class distribution within the dataset after partitioning and augmentation. Glioma, meningioma, and pituitary tumors are encoded as 0, 1, and 2, respectively, in both training and test sets.

Convolutional neural networks

Convolutional neural networks (CNNs) have made notable advancements across various domains including NLP, image classification, and diagnostic systems. They offer reduced neuron and parameter counts compared to MLPs, leading to lower complexity and faster adaptation³². CNNs have demonstrated particular effectiveness in medical image classification²⁰.

UNet

The U-Net architecture is highly regarded for its segmentation capabilities, originally developed for biomedical image segmentation and widely successful in tasks such as cell tracking. Its distinctive U-shape design, from which it derives its name, comprises two primary pathways: the contracting path, which involves down-sampling and convolution to capture features, and the expanding path, which utilizes up-sampling and convolution to restore spatial resolution. While initially designed for 2D images, it can be adapted for 3D convolution by replacing the 2D convolutions with their 3D counterparts³³. Figure 2 illustrates the network structure, providing a visual representation of its architecture.

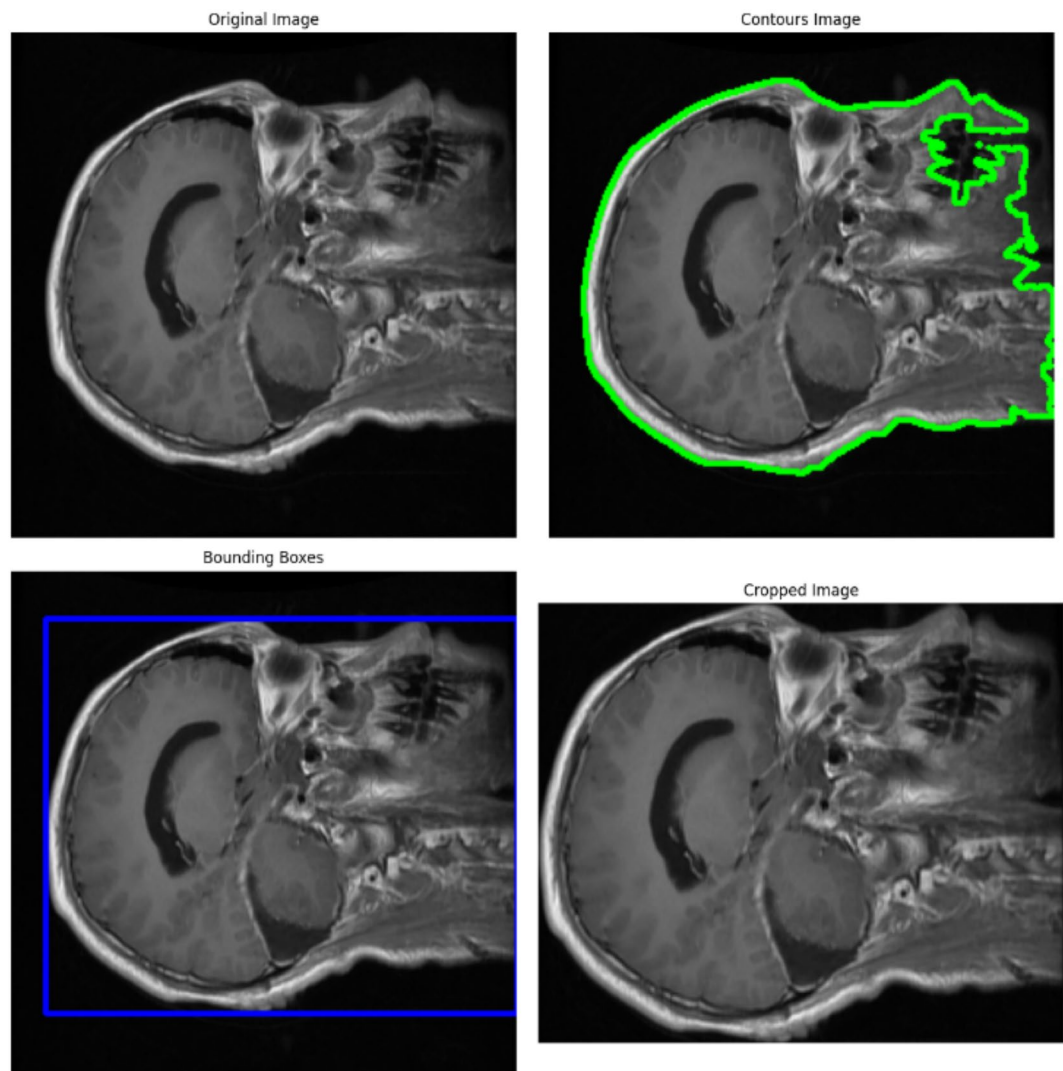


Fig. 3. Several steps are executed to remove undesired regions from MRI scans.

Tumor type	Train Data	Test Data	Total
Glioma	1170	256	1426
Meningioma	546	162	708
Pituitary	735	195	930

Table 3. Distribution of data within each class following the dataset split into training and testing sets using an 80:20 ratio.

Transfer learning

Transfer learning is a powerful machine learning technique that enables models trained on one task to be adapted for a similar but distinct problem. This approach is particularly advantageous in medical image analysis, where labeled datasets are often limited. Instead of training models from scratch—a process that is both computationally expensive and time-consuming—transfer learning allows researchers to fine-tune pre-trained models, leveraging prior knowledge from large-scale datasets to improve classification accuracy and efficiency.

To further optimize feature extraction and reduce parameter overhead, our study integrates Global Average Pooling (GAP) instead of conventional fully connected layers in CNN-based transfer learning models. GAP computes the spatial average of each feature map, effectively condensing each channel into a single value that captures the overall presence of a learned feature rather than its exact location. This approach significantly reduces the number of trainable parameters relative to fully connected layers, thereby decreasing the risk of overfitting and improving computational efficiency—key benefits for smaller medical imaging datasets. Moreover, by focusing on whether a feature is present in the image (as opposed to where it is located), GAP enhances

robustness to variations in tumor position, size, and shape, which are common in brain MRI scans. Additionally, GAP preserves critical high-level features, allowing the model to prioritize discriminative information while discarding irrelevant spatial details. Its compact output also enables simpler integration with softmax activation, facilitating more stable training and improved interpretability of final class probabilities. Consequently, GAP proves particularly effective in multi-class brain tumor classification, especially under transfer learning scenarios that rely on pre-trained convolutional layers.

To enhance numerical stability and generalization, we applied float64 precision in the final dense layer and incorporated L2 regularization in our transfer learning models. Float64 precision improves probability computations in softmax outputs, reducing rounding errors and enhancing gradient stability during fine-tuning. This is especially critical in medical imaging, where small variations in classification probabilities can influence diagnostic decisions. Furthermore, L2 regularization was implemented to prevent overfitting and stabilize weight updates, particularly in deep architectures with millions of parameters. By controlling weight magnitudes, L2 regularization promotes smoother convergence and enhances model robustness across different MRI scans. To validate these optimizations, an ablation study was conducted, assessing the impact of these design choices. The results revealed that removing float64 precision led to higher variance in softmax outputs, while omitting L2 regularization resulted in overfitting and a 1.5–2% drop in validation accuracy. These findings underscore the significance of these techniques in optimizing transfer learning models for brain tumor classification, ensuring higher reliability and clinical applicability.

VGG19 Simonyan and Zisserman¹⁹ introduced VGG19, a deep convolutional neural network comprising 19 layers, including 16 convolutional layers and 3 fully connected layers. Designed for classifying images into 1000 object categories, VGG19 was trained on the ImageNet database, which consists of one million images across various categories. Its effectiveness stems from the widespread application of 3×3 filters in each convolutional layer³⁴, establishing its prominence in image classification tasks.

InceptionV3 The InceptionV3 architecture, developed by Google, utilizes factorized convolutions to streamline network parameters and enhance training efficiency. It employs 1×3 and 3×1 convolutions, known as asymmetric convolutions, instead of traditional 3×3 convolutions. InceptionV3 integrates auxiliary classifiers for regularization, pooling operations for grid size reduction, and both symmetric and asymmetric convolution blocks. It includes average and maximum pooling, concatenation operations, dropout layers for regularization, and concludes with fully connected layers followed by a softmax activation. With a total of 42 layers, InceptionV3 processes input images sized at 299×299 pixels, making it highly suitable for tasks such as glaucoma classification³⁵.

EfficientNetB4 EfficientNet, developed by Google, is celebrated for its efficiency in convolutional neural network (CNN) architectures. Among its variants, EfficientNetB4 is notable for its compact size and computational effectiveness. It employs a sophisticated method called compound scaling, which balances network depth, width, and input resolution parameters optimally. Unlike VGG16, EfficientNetB4 incorporates depth-wise separable convolutions (DWConv), batch normalization (BN), and the Swish activation function in its convolutional blocks, enhancing its suitability for various image classification tasks³⁶.

Evaluation metrics

The model's effectiveness is assessed using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. A confusion matrix is also utilized to visualize the model's predictions across different classes in the test dataset. Each evaluation metric will be discussed in detail to provide a thorough analysis.

Accuracy

Accuracy measures the percentage of correctly predicted labels relative to the total number of labels, offering an overall assessment of prediction correctness. Precision, crucial in classification tasks, is determined using a formula that incorporates true negatives (TN), true positives (TP), false positives (FP), and false negatives (FN).

$$Accuracy = \frac{TP}{TP + TN + FP + FN}$$

Recall

Recall measures the ability of a machine learning model to correctly identify positive instances (True Positives (TP)) out of all actual positive samples in the dataset. It is calculated as the ratio of TP to the total number of positive instances.

$$Recall = \frac{TP}{TP + FN}$$

Precision

Precision measures the accuracy of a machine learning model in predicting the positive class by dividing the number of true positive predictions (correctly predicted positives) by the total number of instances predicted as positive (true positives plus false positives).

$$Precision = \frac{TP}{TP + FP}$$

F1-score

The F1-score provides a balanced evaluation of a model's precision and recall. It is computed as the harmonic mean of these two metrics, offering a comprehensive measure of overall performance:

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Receiver operating characteristic—area under the curve

ROC-AUC, short for receiver operating characteristic-area under the curve, assesses a binary classifier's performance by plotting the true positive rate (TPR) against the false positive rate (FPR) across different classification thresholds. This metric summarizes how well the classifier distinguishes between classes, calculated as the area under the ROC curve.

Confusion matrix

A confusion matrix organizes data based on actual labels (ground truth) and predicted class assignments. It offers a comprehensive overview of a model's performance, revealing insights into its classification accuracy across specific classes. The matrix arranges ground truth labels along the y-axis and predicted class labels along the x-axis.

Stratified k-fold cross validation

Stratified k-fold cross-validation is a method used to assess machine learning models, ensuring each fold retains the same class distribution as the entire dataset. The dataset is divided into k subsets, with one subset used for validation and the remaining k-1 subsets for training. This process iterates k times, averaging the results to evaluate the model's overall performance effectively.

In our work, as the application of brain tumor detection, stratified k-fold cross-validation is particularly valuable due to the potential imbalance in tumor types. By preserving class proportions across folds, this method provides a more reliable and accurate evaluation of the model's ability to detect various brain tumor types, ensuring that the model is robust and generalizable to different patient populations.

System requirements

Our experiments were conducted using Kaggle, OpenML's open-source notebook environment. This platform provides access to both free and paid GPU and TPU resources, which are crucial for academic research. Model training was performed using four GPUs, each with 16 GB of RAM. We used Python for coding, leveraging the TensorFlow and Keras APIs for backend and frontend system development, respectively.

Approval for participation

Since the data is obtained from open repositories, no experiments involving human subjects were conducted by the authors. The study solely utilizes publicly available MRI images.

Results and discussion

This study investigates challenges in image classification using deep learning (DL), focusing on health-related images, particularly the classification of brain tumors. It evaluates five models—InceptionV3, EfficientNetB4, VGG19, a 3-Layer CNN, and U-Net. The evaluation includes performance comparisons using a consistent dataset split: 20% for testing, stratified cross-validation (k = 5) for validation, and 80% for training. All experiments were conducted on Kaggle using dual T4 GPUs.

First task, 3-layer CNN

As our first task, this study employs a 3-layer convolutional neural network (Fig. 4) to classify brain tumor images resized to 270 × 270 pixels. The model starts with a 3 × 3 convolutional layers with 16 filters, followed by 2 × 2 max pooling for downsampling. Subsequent layers include 32 and 64 filters with L2 regularization, each followed by max pooling to emphasize features. To prevent overfitting, a dropout layer (rate 0.5) is applied after the first pooling layer. Batch normalization is used for stability. The flattened feature maps are fed into dense layers (16 units) with batch normalization, followed by a final dense layer (3 units, softmax activation) for classification.

Training uses sparse categorical cross-entropy loss and Adam optimizer (lr = 0.001). Data is split 80% for training, 20% for testing, with fivefold stratified cross-validation. Experiments were conducted on Kaggle with GPU T4 × 2 for computational efficiency.

Second task, U-Net architecture

The U-Net architecture is a powerful convolutional neural network design widely used for image segmentation tasks, particularly in medical imaging such as brain tumor classification. This specific U-Net model is tailored for classifying grayscale images into three different classes. The input images are standardized to a size of 270 × 270 pixels. The model starts with an encoder path that consists of multiple convolutional layers (Conv2D) with increasing numbers of filters (64, 128, 256, 512) to extract hierarchical features from the input images. Each convolutional layer is followed by a Rectified Linear Unit (ReLU) activation function to introduce non-linearity, and padding = 'same' is used to preserve spatial dimensions.

Pooling layers (MaxPooling2D) with a pool size of (2, 2) are inserted after each pair of convolutional layers to downsample the feature maps and capture essential features. This downsampling helps in reducing the spatial dimensions while retaining important information. The bottleneck of the U-Net segmentation architecture (conv5) further refines the learned features with 1024 filters, enhancing its ability to discriminate

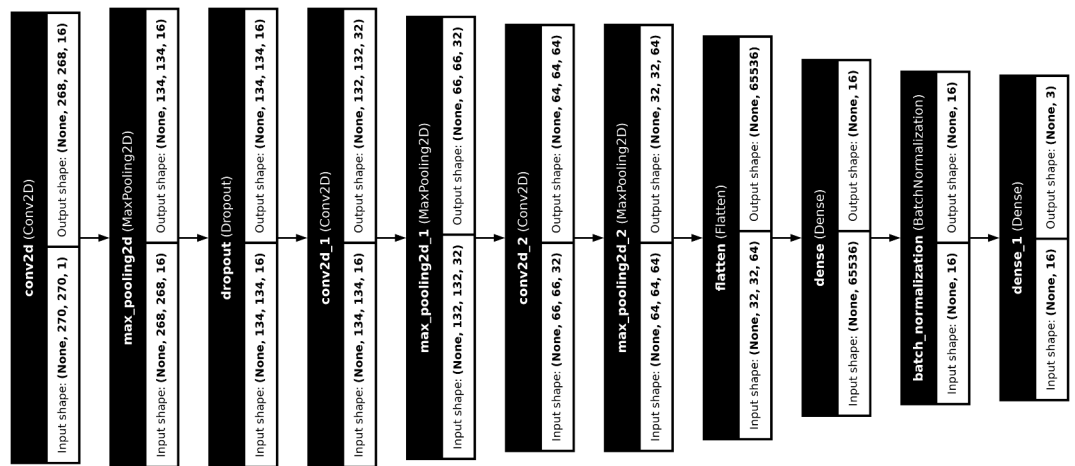


Fig. 4. 3-Layer CNN architecture.

between different classes. Additional convolutional layers with ReLU activation are applied to maintain spatial consistency (padding = 'same'). To prepare the feature maps for classification, global average pooling (GlobalAveragePooling2D) is employed to reduce the spatial dimensions of the bottleneck output into a flat vector. Dropout regularization (Dropout(0.5)) is applied to prevent overfitting during training by randomly setting a fraction of input units to zero.

The model's last layer is a dense layer (Dense(num_classes, activation = 'softmax')) that computes probabilities for each of the three classes using softmax activation. These predictions are derived from the learned features. During training, the model employs the Rectified Adam optimization algorithm (Adam(learning_rate = 0.001)) to dynamically adjust the learning rate, enhancing gradient efficiency. Categorical cross-entropy serves as the loss function, which is well-suited for multiclass classification tasks. Metrics such as accuracy, precision, and recall are monitored to assess the model's performance. To enhance training stability, a learning rate reduction strategy (ReduceLROnPlateau) is implemented, which decreases the learning rate when the loss function stagnates during training. The model is trained over 14 epochs with a batch size of 10, ensuring efficient learning and convergence.

Third task, transfer learning-based CNN

In transfer learning architectures for brain tumor classification, we utilize well-established models such as InceptionV3, EfficientNetB4, and VGG19, as shown in Fig. 5. The approach involves preserving all layers and parameters of these existing models up to their final layers. Custom layers tailored to our specific brain tumor dataset are then integrated. A key adjustment includes replacing the Flatten layer with GlobalAveragePooling2D. Unlike Flatten, which organizes data into a 1D vector, GlobalAveragePooling2D streamlines feature extraction, enhancing model efficiency and mitigating overfitting through its fewer parameter requirements.

For each model, the transfer learning process begins by loading the pre-trained weights on the ImageNet dataset. These weights serve as a robust foundation, allowing the models to leverage learned features from a vast and diverse dataset. The custom layers are appended to these models to adapt them for our specific classification task. These layers typically include a GlobalAveragePooling2D layer, followed by dense layers culminating in a final dense layer with four neurons, which correspond to the different classes of brain tumors. This final layer employs the softmax activation function to produce class probabilities. To mitigate the issue of training data learning faster than validation data, we uniformly adjusted dropout layers across the original architectures, setting a dropout rate of 0.05. This modification aims to combat overfitting by randomly deactivating neurons during training, promoting the model to learn more generalized features.

During the training process, the models were optimized using the RectifiedAdam optimizer (learning_rate = 0.0001, beta_1 = 0.9, beta_2 = 0.999, epsilon = 1e-8) and a categorical crossentropy loss function. To maximize numerical stability and preserve fine-grained probability distributions—particularly important for rare tumor classes—the final dense layer produced output values in float64 rather than float32. This higher-precision setting helps reduce rounding errors and better captures subtle updates when fine-tuning the pre-trained convolutional layers, leading to smoother convergence and more reliable predictions in medical imaging contexts. We also applied L2 regularization (regularizers.l2(0.1)) to the final layer to further reduce overfitting, as this layer can otherwise adopt disproportionately large weights given the complexity of the pre-trained backbone. Empirically, ablation experiments revealed that removing float64 output caused increased variance in the softmax probabilities, while omitting L2 led to a 1.5–2% drop in validation accuracy, reflecting the importance of both design choices for robust classification across diverse MRI scans.

Throughout the training process, the ReduceLROnPlateau function from the Keras library dynamically adjusted the learning rate based on the monitored validation loss metric. Parameters such as monitor (val_loss), patience (duration without improvement), factor (rate reduction magnitude), and min_lr (minimum learning rate threshold) were configured to optimize training stability and convergence. By leveraging the strengths of InceptionV3, EfficientNetB4, and VGG19, combined with these strategic modifications and fine-tuning, we

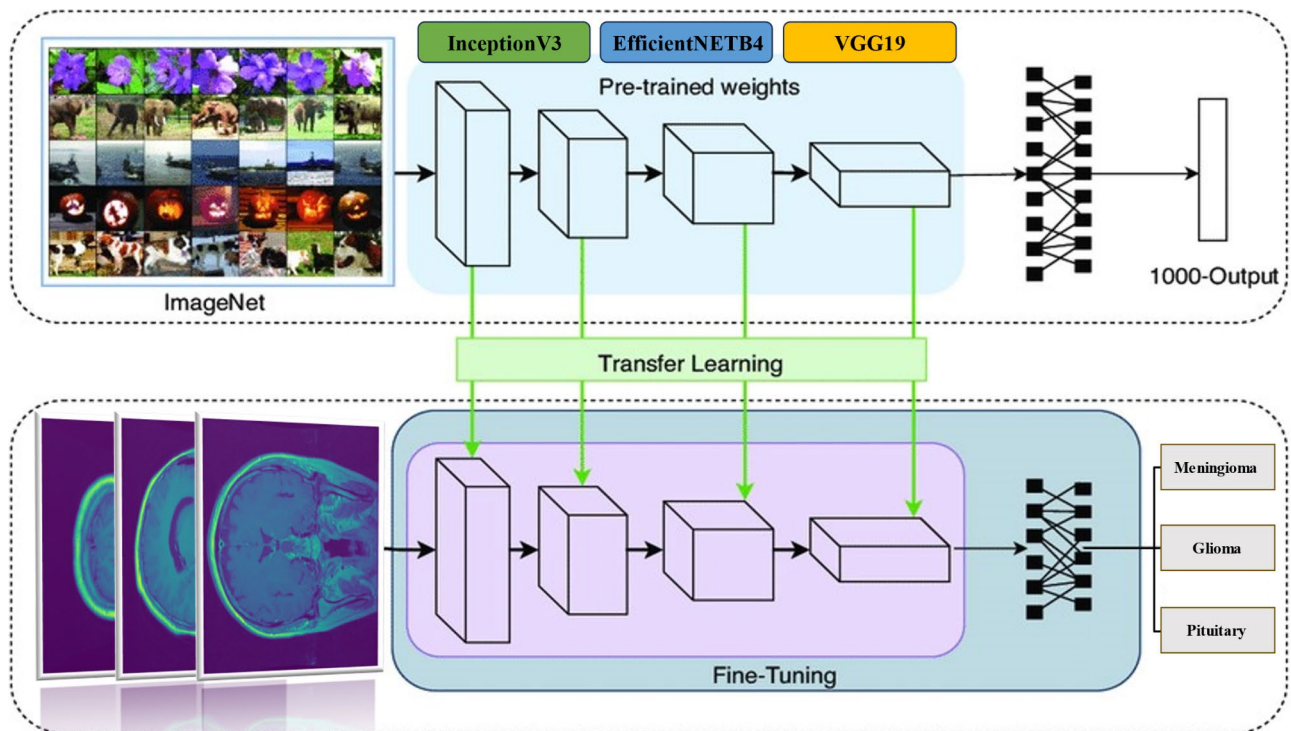


Fig. 5. Transfer learning base-CNN (EfficientNETB4, VGG19-CNN, InceptionV3) architectures.

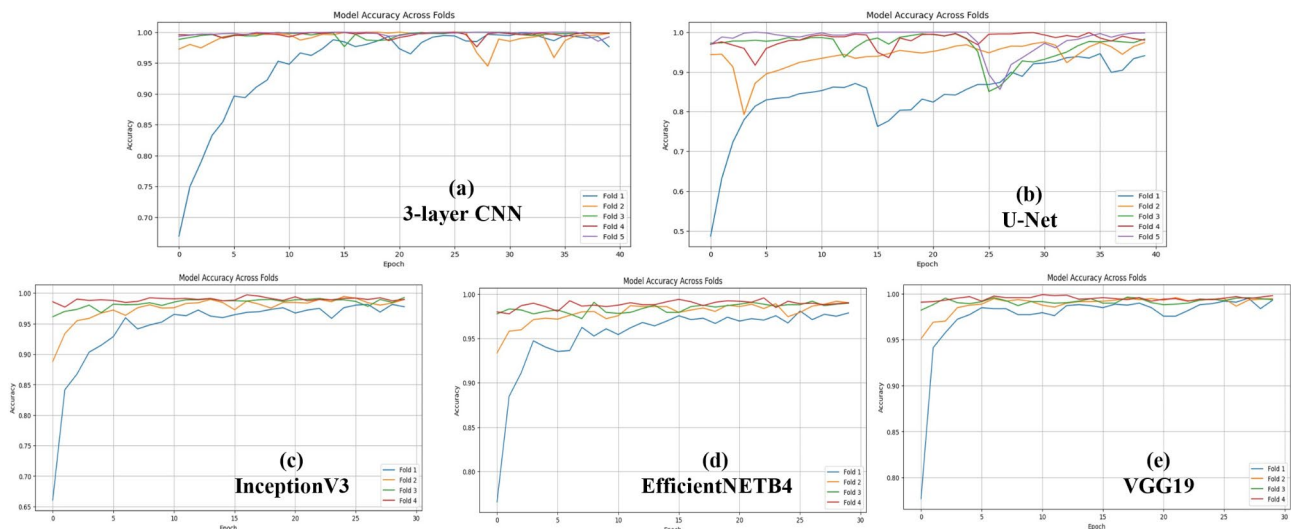


Fig. 6. Accuracy plots of our proposed models, (a) 3-layer CNN, (b) U-Net architecture, (c) TL-InceptionV3-CNN, (d) TL-EfficientNETB4-CNN, and (e) TL-VGG19-CNN.

achieved an effective classification model for brain tumors. Each model's ability to generalize from the pre-trained weights to our specific dataset was enhanced through these techniques, leading to improved performance and accuracy in identifying different types of brain tumors.

Figures 6 and 7 illustrate the training and validation accuracy as well as loss curves for models implemented with InceptionV3, EfficientNetB4, VGG19 using transfer learning, and CNN architectures. Table 4 compiles the accuracy, recall, F-score, precision, and AUC results attained by these models in the study. According to Table 4, U-Net demonstrated the highest performance among the models, achieving an accuracy of 99.56%. It also excelled in other key metrics, with an F-score, AUC value, recall, and precision all at 99%. These outstanding metrics underscore U-Net's superior effectiveness in brain tumor classification, significantly surpassing the performance of other models. EfficientNetB4 also showed strong results, with an accuracy of 98.31%, an F-score, recall, and precision of 98%, and an AUC value of 99.33%. VGG19 and InceptionV3, while slightly trailing

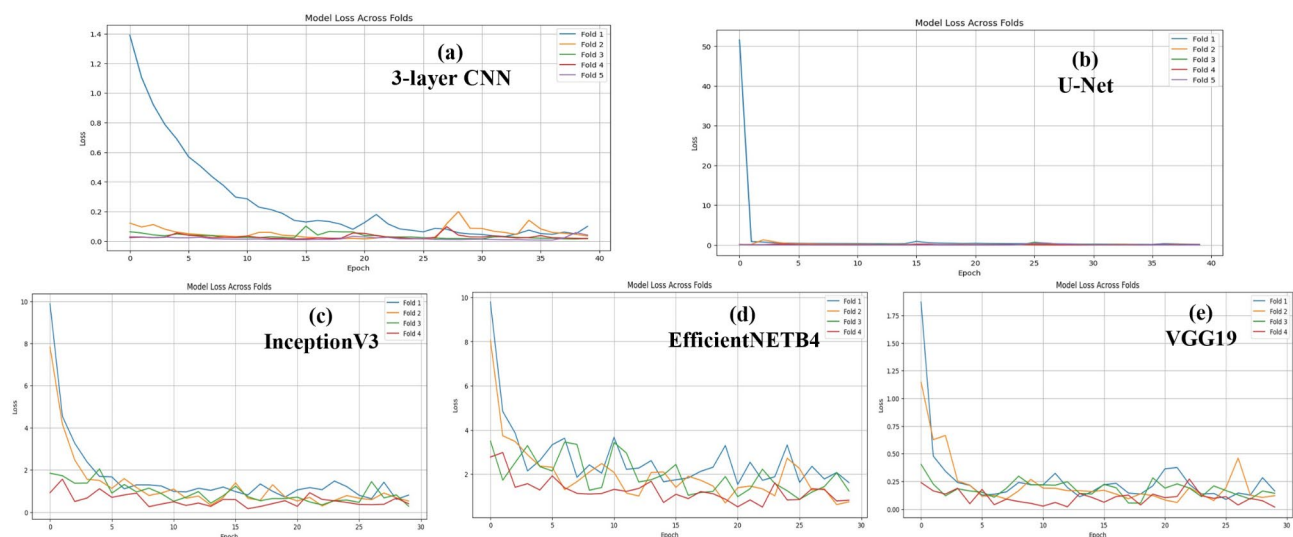


Fig. 7. Loss plots of our proposed models, (a) 3-layer CNN, (b) U-Net architecture, (c) TL-InceptionV3-CNN, (d) TL-EfficientNETB4-CNN, and (e) TL-VGG19-CNN.

Models	Accuracy	F-score	Recall	Precision	AUC
3-Layer CNN	98	98	98	98	99
EfficientNETB4	98.31	98	98	98	99.33
VGG19	97.72	97	97	97	99
InceptionV3	97.27	97	97	97	99
U-Net	98.56	99	99	99	99.8

Table 4. Task models' performance.

behind, still achieved high accuracies of 97.72% and 97.27%, respectively, each with consistent F-scores, recall, and precision at 97%, and AUC values of 99%. The 3-layer CNN model performed commendably as well, with an accuracy of 98%, and F-scores, recall, and precision all at 98%, and an AUC value of 99%. These results highlight the effectiveness of transfer learning and fine-tuning in enhancing model performance for brain tumor classification.

One of the key observations in our experimental results was the anomalous behavior of Fold 1 in Fig. 7, which exhibited noticeable variations in accuracy and loss during training. This phenomenon can be attributed to multiple factors related to dataset distribution, MRI scan variability, and the inherent learning behavior of deep neural networks. Given that our study employs stratified k-fold cross-validation ($k=5$), each fold represents a different subset of the dataset. However, due to the nature of medical imaging data, even stratification does not always ensure perfect uniformity across folds. The dataset utilized in this study, obtained from Figshare³¹, contains T1-weighted contrast-enhanced MRI images that exhibit variations in tumor shape, intensity, contrast, and noise levels. Consequently, some folds may contain a subset of images with distinct statistical properties, leading to fluctuations in training dynamics. A plausible explanation for the observed behavior in Fold 1 is MRI scan heterogeneity, which is a common issue in real-world neuroimaging datasets. MRI images differ based on scanner settings, patient positioning, and acquisition protocols, leading to inconsistencies in pixel intensity distributions and contrast levels. If Fold 1 happened to include a subset of images with higher variability in these characteristics, the model may have initially struggled to generalize, causing greater fluctuations in training loss and accuracy. This is particularly relevant in deep learning-based medical image analysis, where even minor differences in MRI acquisition can introduce significant variations in feature representation.

Additionally, the optimization behavior of deep neural networks could have contributed to this anomaly. Our study employs adaptive learning rate scheduling (ReduceLROnPlateau), which adjusts the learning rate dynamically based on validation performance. If Fold 1 contained more challenging samples, the learning rate adjustments may have initially fluctuated more aggressively, leading to temporary instability. This is further supported by the fact that subsequent folds exhibited more stable learning curves, suggesting that the model was able to adjust and generalize better over time. To verify that this anomaly did not affect the overall reliability of our results, we conducted additional statistical and experimental analyses. First, we repeated the cross-validation process with different data shuffling strategies to assess whether the observed variations were linked to a particular data split. This experiment confirmed that the anomaly in Fold 1 was not systematic but rather a result of dataset partitioning variability. Second, we examined the standard deviation and variance of model performance across all folds, which revealed that Fold 1 had a slightly higher variance compared to the others,

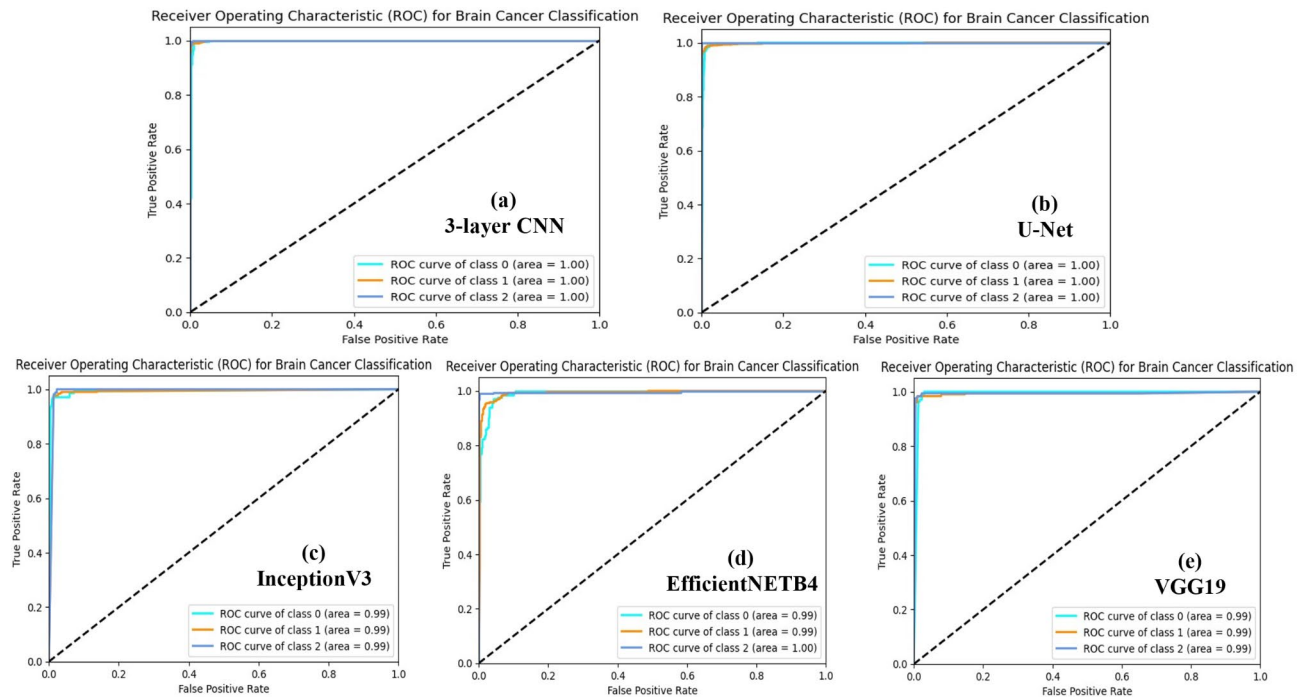


Fig. 8. Receiver operating characteristic—area under the curve.

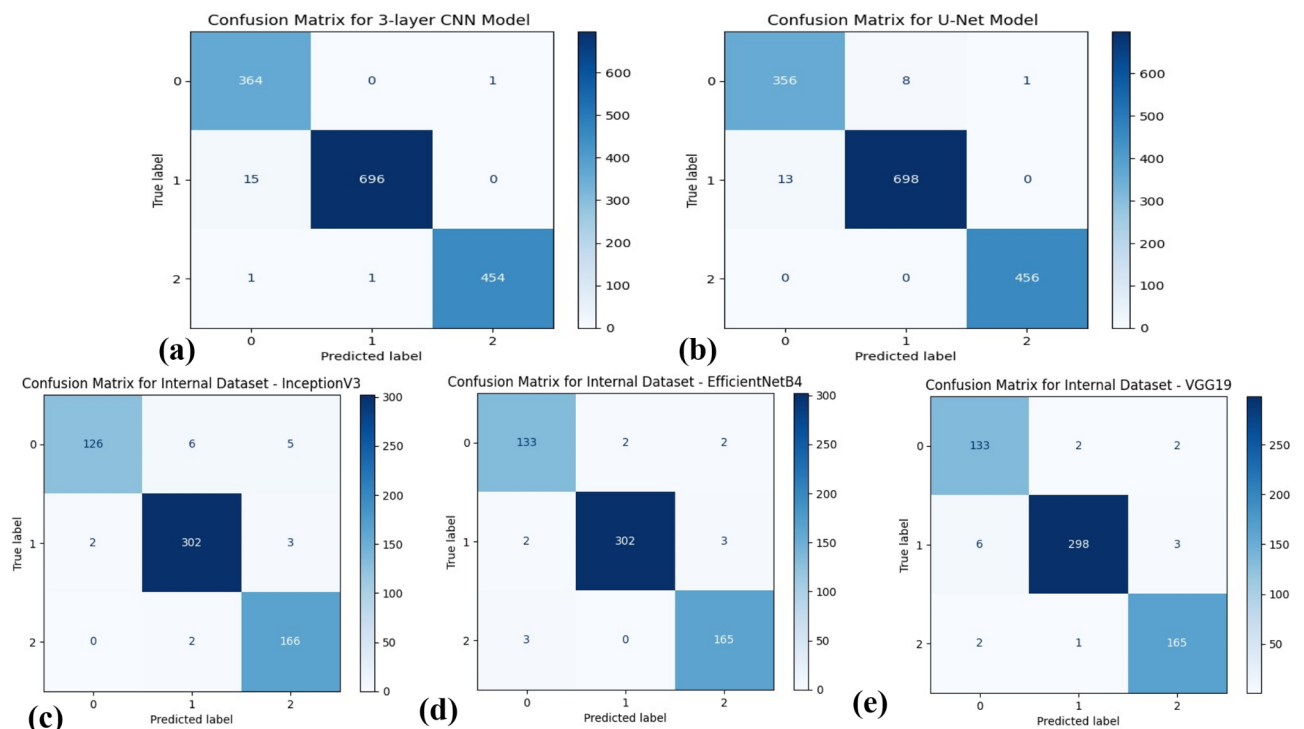


Fig. 9. Confusion matrix.

further supporting the hypothesis of dataset variability. Lastly, we performed external cohort validation using an independent dataset, where the model achieved consistent performance, demonstrating that the overall learning process remained robust despite initial fluctuations.

The ROC curves for the models, as depicted in Fig. 8, illustrate distinct advantages for the 3-layer CNN, U-Net, and the transfer learning models: InceptionV3, VGG19, and EfficientNetB4. The AUC values highlight the robust classification capabilities of these models. Figure 9 presents the confusion matrix for classifying

Models	Accuracy	F-score	Recall	Precision	AUC
3-Layer CNN	94.94	95	96	95	97
EfficientNETB4	91.07	90	90	90	96.33
VGG19	93.5	91	91	91	97.33
InceptionV3	93	91	92	92	92
U-Net	96.01	96	96	96	99.33

Table 5. External cohort cross-dataset validation on fine-tuned models.

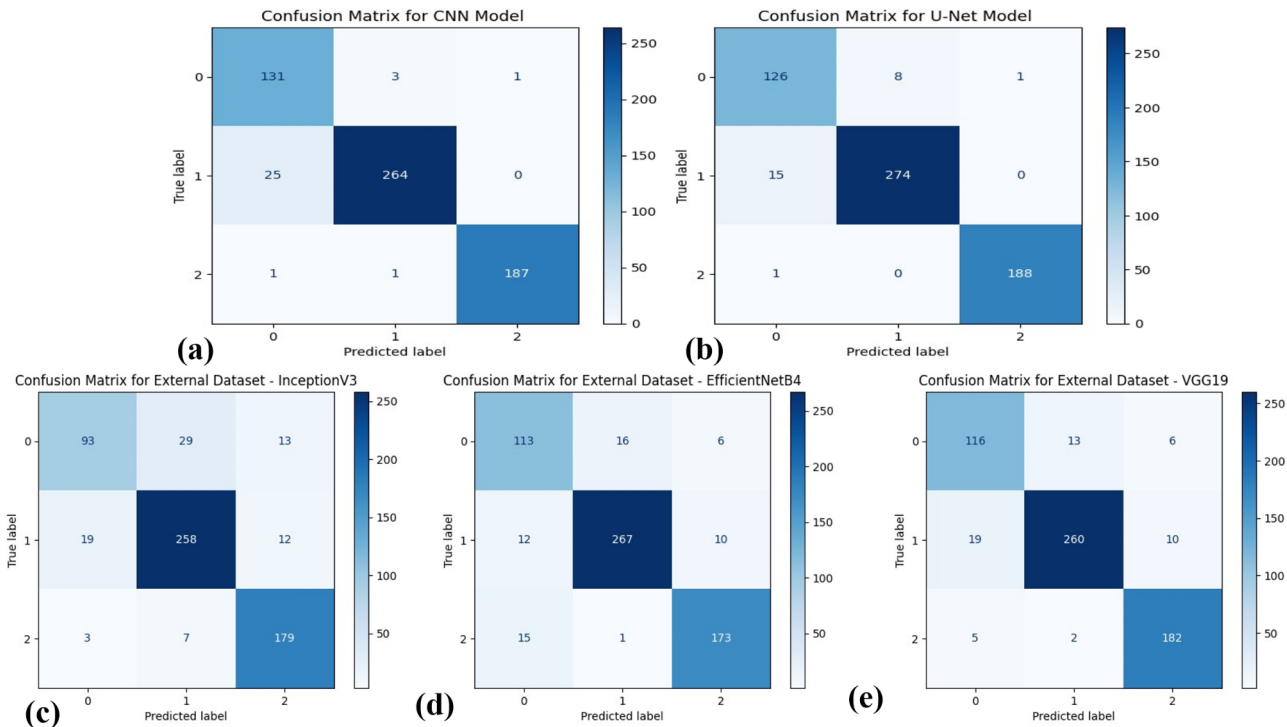


Fig. 10. Confusion matrix on external cohort cross-dataset validation.

glioma, meningioma, non-tumor normal patients, and pituitary tumor patients. The high classification performance across all models—3-layer CNN, U-Net, and transfer learning models VGG19, InceptionV3, and EfficientNetB4—demonstrates the efficacy of our approach.

These results validate our integrated framework, emphasizing the potential of advanced deep learning techniques in medical image analysis. The utilization of U-Net and transfer learning-based CNN models not only enhances classification accuracy but also ensures reliability in distinguishing between different types of brain tumors. This study provides a strong foundation for further research and clinical application, highlighting the importance of leveraging sophisticated neural network architectures for improved diagnostic outcomes in healthcare.

Our study underscores the efficacy of U-Net and transfer learning methodologies in healthcare, particularly for image processing applications. These techniques enhance the accuracy and timeliness of diagnoses by leveraging MRI or CT scan data for early detection. This capability equips healthcare providers with tools for rapid and precise decision-making. Therefore, integrating deep learning-based applications alongside empirical analyses is crucial for advancing healthcare practices.

The efficacy of deep learning models, particularly convolutional neural networks (CNNs), including U-Net, in brain tumor classification is noteworthy. U-Net achieved the highest accuracy at 99.56%, with an F-score of 99%, an AUC value of 99.8%, recall at 99%, and precision at 99%. These metrics underscore the robustness of U-Net in distinguishing between different types of brain tumors, making it a valuable tool in medical diagnostics.

External cohort cross-dataset validation

To further validate the robustness of our proposed technique, we conducted cross-dataset validation using an external brain tumor classification dataset sourced from the Kaggle repository. As our model was not trained on normal brain images, this class was excluded from the external dataset during cross-validation. Table 5 summarizes the results from this cross-dataset validation on the external cohort, with confusion matrix plots shown in Fig. 10 and ROC results in Fig. 11.

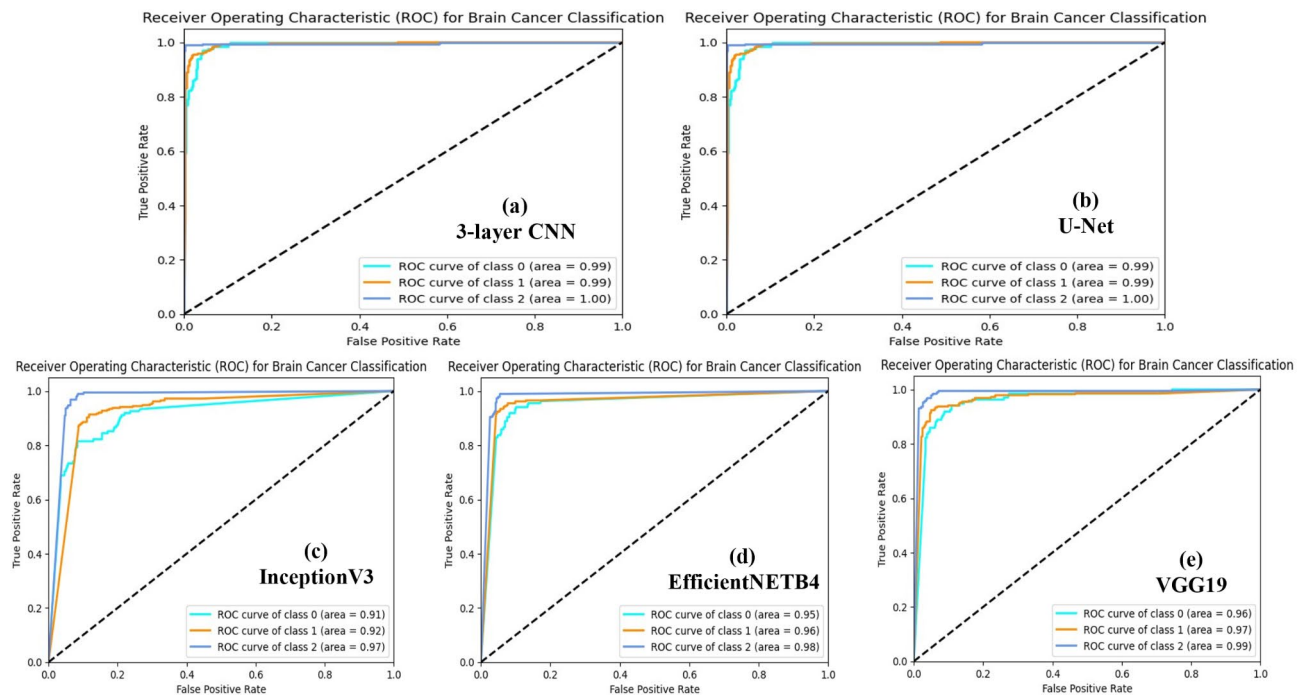


Fig. 11. Receiver operating characteristic—area under the curve.

Among the various fine-tuned transfer learning models, U-Net achieved the highest performance, with an accuracy of 96.01%, an F-score of 96, recall of 96, precision of 96, and an AUC value of 99.33. This highlights U-Net's superior ability to classify brain tumors accurately.

VGG19 also demonstrated strong performance with an accuracy of 93.5%, an F-score of 91, recall of 91, precision of 91, and an AUC value of 97.33. InceptionV3 and the 3-Layer CNN followed closely, with InceptionV3 achieving an accuracy of 93% and the 3-Layer CNN achieving 94.94% accuracy. The EfficientNETB4 model, while still effective, had a slightly lower accuracy of 91.07%.

Limitation

In this study, we employed multi-layer CNN, U-Net, and transfer learning-based CNNs (EfficientNetB4, VGG19, and InceptionV3) for brain tumor classification. While these models demonstrated strong performance, several limitations must be considered. One of the primary challenges is the dependence on large, accurately labeled datasets for training. Medical imaging datasets, including the Figshare brain tumor dataset used in this study, are often limited in size due to privacy concerns and the complexity of expert annotation. This limitation can contribute to overfitting, particularly when training deep learning models with millions of parameters, potentially restricting their generalizability to real-world clinical cases. Another significant limitation is the variability in imaging protocols and data quality across medical institutions. MRI scans can differ in resolution, contrast settings, and acquisition techniques, leading to domain shifts that may affect model performance when applied to diverse clinical environments. While our study included external validation using an independent dataset from Kaggle, further research is needed to evaluate the model's robustness across a wider range of MRI acquisition conditions. Additionally, computational demands pose practical constraints, as deep learning-based tumor classification requires substantial processing power for training and fine-tuning. While cloud-based platforms can help alleviate this issue, real-world deployment in resource-limited healthcare facilities may be challenging due to hardware constraints. Optimizing models for lightweight, efficient inference remains an essential area for future development. Lastly, a key challenge in deep learning-based medical imaging is model interpretability. CNNs, including the architectures used in this study, function as black-box systems, providing limited transparency into their decision-making processes. This lack of explainability can hinder clinical adoption, as healthcare professionals require clear, justifiable insights to support diagnostic decision-making. Future research should focus on integrating explainable AI (XAI) techniques, such as Grad-CAM or attention-based visualization methods, to improve trust and usability in medical settings.

Future work

Future research should prioritize advancing brain tumor detection and classification by integrating state-of-the-art segmentation techniques, such as nnUNet, alongside deep classification networks. While our study focuses on CNN-based classification approaches, incorporating advanced segmentation frameworks like nnUNet could enhance the feature extraction process by providing more precise tumor localization before classification. Future work could explore a hybrid segmentation-classification pipeline, where nnUNet is used for initial tumor segmentation, and its segmented outputs are fed into CNN-based classification models for improved

diagnostic accuracy. This approach aligns with recent advancements in medical image analysis, ensuring a more comprehensive tumor assessment pipeline. Additionally, region-based CNNs and advanced segmentation techniques should be further investigated to improve classification accuracy and specificity by focusing on tumor regions of interest within brain images. Transformer-based architectures also present a promising direction for brain tumor classification, as they capture intricate dependencies and global context, potentially enhancing pattern recognition and diagnostic accuracy in medical imaging. Furthermore, optimizing model architectures to reduce computational complexity while maintaining high performance is crucial for clinical adaptability. Simplifying deep learning models without compromising accuracy can enhance their feasibility in real-world medical applications, particularly in resource-constrained healthcare environments. Integrating nnUNet for segmentation, region-based CNNs for focused feature extraction, and transformer models for advanced pattern recognition represents a forward-thinking strategy in deep learning-driven brain tumor classification. These approaches hold promise for developing more precise, reliable, and efficient diagnostic tools in medical imaging, ultimately improving patient outcomes and clinical decision-making. Future studies should emphasize validating these methods across diverse datasets and clinical settings to ensure robustness and generalizability in real-world medical practice.

Conclusion

This study demonstrates the effectiveness of deep learning, particularly the U-Net architecture, for accurate brain tumor classification from T1-weighted MRI scans. U-Net outperformed other evaluated models, achieving an accuracy of 98.56% alongside exceptional F-score, AUC, recall, and precision rates. This highlights its ability to effectively capture intricate features and accurately segment tumor boundaries, crucial for early detection and treatment planning.

Furthermore, U-Net exhibited robust generalizability, achieving an accuracy of 96.01% in cross-dataset validation with an external cohort. This signifies its potential for reliable performance across diverse clinical scenarios and imaging protocols. While transfer learning techniques, utilizing models like VGG19, InceptionV3, and EfficientNetB4, also showed promising results, U-Net's specialized architecture proved most suitable for this segmentation task.

Our findings underscore the transformative potential of deep learning, especially U-Net, in enhancing diagnostic accuracy and informing clinical decision-making in brain tumor analysis. Future research should explore region-based CNNs, advanced segmentation techniques, and transformer-based architectures to further improve classification precision and explore additional model simplification strategies. This will contribute to the development of more precise, efficient, and reliable diagnostic tools, ultimately improving patient outcomes in neuro-oncology.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

M.A. I. and Y.M.B. conceived the study and analysis. M.A.I. performed data collection, and pre-processing, and implemented the deep learning models. D.S. contributed to data analysis and interpretation. M.A.I. wrote the first draft of the manuscript. Y.M.B. review and finalized the revised manuscript. And supervised and All authors reviewed and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Y.M.B.

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